BELOW, ONLY

1	2	3	4	5	6	7	8

1. **The following is an example of an $O(1)$ time complexity algorithm:**

   A. Accessing the weight of an edge in an adjacency matrix representation
   B. Finding largest/smallest element in a graph.
   C. Determining if a specific item is included in the optimal solution for using the traceback method.
   D. Calculating the minimum cost of a path in a weighted graph.

2. **What is the closest time complexity of the function $f(n) = n^2 + log\,(n)$**

   A. $O\,(n \log n)$
   B. $O\,(nlog\,(n)^2)$
   C. $O\,(n^2)$
   D. $O\,(n^2 + nlog\,(n)^2)$

3. **Which of the following expressions grows the fastest as $n \to \infty$ (worst eff**

   A. $\sqrt{n}\,log(n)$
   B. $\sqrt[3]{n}\,log(n)$
   C. $\sqrt[3]{n}\;log^2(n)$
   D. $\sqrt{n}\;log^2(n) + \sqrt[3]{n}log\,(n)$

4. **The solution of this recurrence $T(n) = 9\,T(n/3) + n^2$ is (*Hint: use M***

   A. $O\,(n^2)$
   B. $O\,(n^3)$
   C. $O\,(n^2 \log n)$
   D. Cannot be solved using Master Theorem

5. **After applying Build-Max-Heap on the array [7, 2, 5, 8, 4], the resultin**

   A. [8,7,5,2,4]
   B. [8,5,7,2,4]
   C. [7,8,5,2,4]

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6. **Which of the following algorithms is designed without using a queue in its implementation?**

   A. Bellman-Ford
   B. Breadth-First Search (BFS)
   C. Dijkstra's Algorithm
   D. Prim's Algorithm

7. **Consider the following list of fruits:**

   apple, banana, kiwi, peach, grape, pear, melon, plum

   If Radix Sort is applied, sorting based on letters starting from the least significant character (rightmost) to the most significant character (leftmost), how many passes will be required to fully sort the list?

   A. 4 passes
   B. 5 passes
   C. 6 passes
   D. 7 passes

8. **How many comparisons are made in the Counting Sort algorithm when sorting n elements in the range $[0, k]$?**

   A. $O(n + k)$
   B. $O(n\ log k)$
   C. $O(k\ log\ n)$
   D. None of the above

9. **You are tasked with distributing tasks among workers to minimize the maximum workload assigned to any worker. Each task has a processing time, and you can assign tasks to workers in any order. The goal is to balance the load while ensuring all tasks are completed.**

   Which algorithm is the best suited to solve this problem?

   A. Dynamic Programming
   B. Greedy Algorithm
   C. Bellman-Ford Algorithm
   D. Prim's Algorithm

10. **You are tasked with designing a communication network between multiple cities. Each city must directly or indirectly connected, and the cost of laying communication cables between any two ci is known. The goal is to minimize the total cost of the network.**

    Which algorithm is best suited to solve this problem?

    A. Bellman-Ford Algorithm
    B. Dijkstra's Algorithm
    C. Dynamic Programming
    D. Prim's Algorithm

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Question 2: *NP-complete* To be answered in **(20) Minutes** [ ] / **4 Marks**

A. Match the terms in Column A with their corresponding descriptions in Column B. [0.5 mark for each]

Column A (Concepts)	Write the Appropriate letter in Column B	Column B (Descriptions)
P (Polynomial Time)		A. All problems solvable in polynomial time.
NP (Nondeterministic Polynomial)		B. Hardest problems in NP that are both in NP and NP-Hard.
NP-Complete		C. Problems that can be verified in polynomial time but may not be solvable in polynomial time.
NP-Hard		D. An NP-Complete problem that involves determining the satisfiability of a formula in CNF form.
3-SAT		E. Problems that are at least as hard as NP-Complete problems but may not be in NP.
Polynomial Reduction		F. Approaches to find good but not guaranteed optimal solutions to NP-Complete problems.
Cook-Levin Theorem		G. Established the first NP-Complete problem, Boolean Satisfiability (SAT).
Heuristics		H. Transforming one problem into another in polynomial time to compare complexity.

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Question 3: *Sorting* **To be answered in (35) Minutes** [ ] / 10 M

A. For each of the following scenarios, choose the most suitable sorting algorithm from the following list: **[1 ma for each]**

Merge Sort, Insertion Sort, Heap Sort, Counting Sort, Radix Sort

Scenario	Write the Appropriate Sorting Algorithm
The dataset contains integers within a small, known range (e.g., 1 to 1000).	
You are working with a large dataset where stability is important, and you need $O(nlogn)$ time complexity.	
You have a nearly sorted array with only a few elements out of order.	
You need to sort a large list of integers with varying digit lengths, such as phone numbers or zip codes.	
Memory is a constraint, and you cannot use extra memory for auxiliary arrays.	

B. Given the following array $A = [5,2,1,4,2,1]$, apply **Counting Sort** to sort A. The range of values in the array is 1 to 5.

1. Show the contents of the array and C (the count array) after the counting step. **[2 Mark]**

Array C						

2. Show the contents of the array and C (the count array) after the cumulative sum step. **[2 Mark]**

Array C						

3. What is the time complexity of the Counting Sort algorithm? **[1 Mark]**

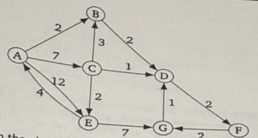
_____

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Question 4: Graph and Dynamic programming — To be answered in (45) Minutes [ ] / 16 Marks

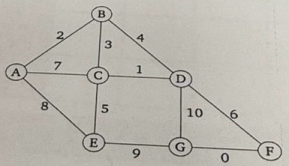
A. Consider the following directed graph. [5 Marks]



Run Dijkstra's algorithm on the above directed graph, starting at vertex A. What is the order in which vertices get removed from the priority queue? Show your answer in the table below

V	A	B	C	D	E	F	G
v.d	0						
v.π	nil						
order	1						

B. Consider the following undirected graph *G*: [5 Marks]



Using the Kruskal Algorithm, find the minimum spanning tree (***MST***) by identifying the edges included in the MST and the value of MCST. Use the table to keep track of sets and selected edges, first row is example run.

Step	Sets	Selected edge	Include/not included in MST
1	{A} {B} {C} {D} {E} {F} {G}	(F,G)	Yes
2			
3			

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6			
7			
8			
9			
10			
11			

Total cost for MST is: _____

C. You are given the following items with their associated benefits and weights, and a knapsack with a capacity W=10kg. Find an **optimal solution** (i.e., a solution with maximum benefit) to fill the knapsack with these items, where items are divisible (i.e., a fraction of an item can be considered). **Note:** If two items have the same benefit relative to their weight, choose the one with the **lower weight** first. **[2 Marks]**

Item (i)	1	2	3	4
Benefit (b_i)	10	15	40	25
Weight (w_i) in kg	1	3	4	5

Answer:

Item (i)	1	2	3	4
Benefit (b_i)				
Weight (w_i)				
b_i/w_i				

Knapsack weight	Items in knapsack	Cost

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Consider the following 0/1 knapsack problem with four items:

Item (i)	1	2	3	4
Benefit (b_i)	10	15	40	25
Weight (w_i) in kg	1	3	4	5

The capacity of the knapsack is **5kg**. Solve the above knapsack problem using dynamic programming. **[4 Marks]**

1. **Fill the table below.**

i \ w	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0					
2	0					
3	0					
4	0					

2. **Which items are included in the knapsack? Show on the above matrix the steps you have followed.**

*The optimal knapsack should contain items:*_____

3. ***The optimal total profit is:***_____

End of Qu